Push-button LED (3 V)

Build a circuit on a half-size breadboard where holding a push button lights a red LED, powered by two AA batteries (3 V).

You will need

- Half-size (400-point) solderless breadboard
- 2×AA battery holder with on/off switch and red/black leads (3 V). Keep the switch OFF while building.
- 1 × red 5 mm LED
- 1 × 100 Ω resistor (brown-black-brown)
- 1 × 6×6 mm 4-pin tactile push button
- Jumper wires (red, black, any other colour)

SAFETY

Battery power only. Never connect anything to mains power.
Keep the battery switch OFF until Step 6.
If anything gets warm, smells or smokes, switch the battery OFF and call the TA.

Step 1: Power rails

Battery switch OFF. Push the red lead into the top + rail (red line) and the black lead into the top − rail (blue line).

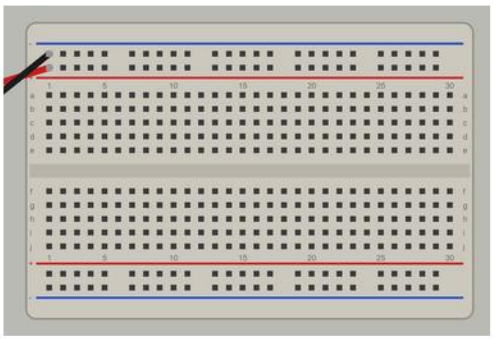


Figure 1: board after step 1

Step 2: Resistor

Put one resistor leg in the top + rail and the other leg in a numbered strip in the top half (e.g. strip 10).

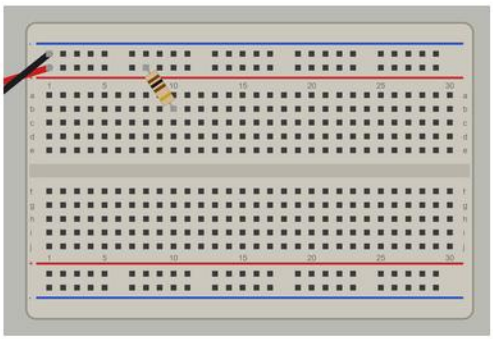


Figure 2: board after step 2

Step 3: LED

Put the LED's long leg (+) in the same strip as the resistor's lower leg, and the short leg (−, flat edge) in the next strip along.

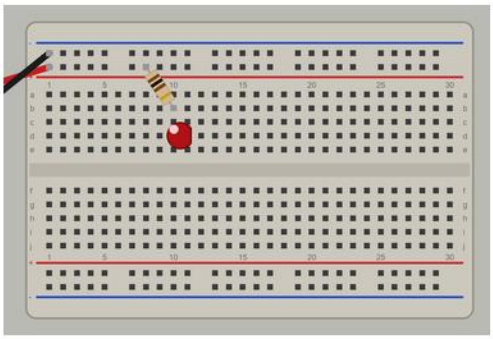


Figure 3: board after step 3

Step 4: Push button

Place the button across the centre channel so one pair of its legs shares the LED's short-leg strip.

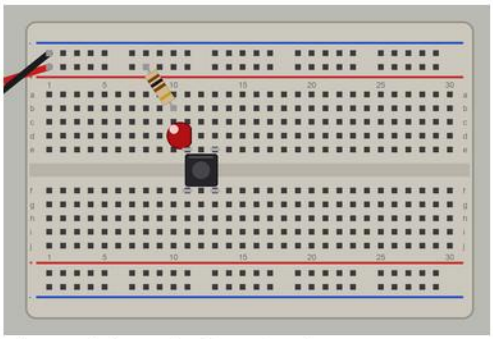


Figure 4: board after step 4

Step 5: Ground wire

Add a black jumper from the button's other leg strip to the top − rail.

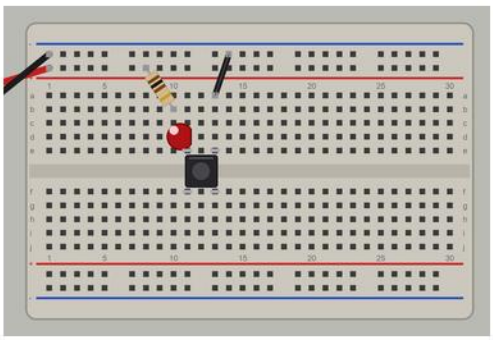


Figure 5: board after step 5

Step 6: Power test

Switch the battery ON. Hold the button down, then press Check.

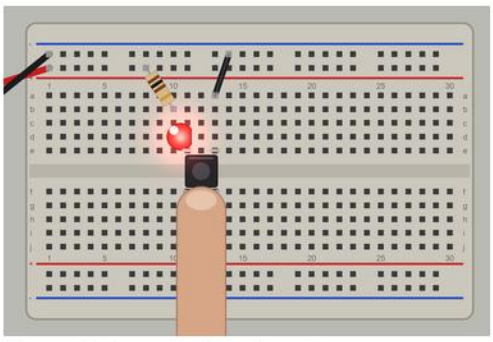


Figure 6: board after step 6

Checkpoint: show your TA the lit LED before you pack up.